

CHM111 Practice Exam on chapter(s): 7, 8

1. Yes $\because$ the de Broglie wavelength $\lambda = \frac{h}{mv}$ is larger than the size of the object.
2. $A \rightarrow F$ $\because$ the most amount of energy is absorbed in this transition and energy is inversely related to wavelength.
3. $E \rightarrow A$ $\because$ the most amount of energy is emitted in this transition and energy is directly related to frequency.
4. Emitted $\because$ a photon is emitted when an electron transitions to a lower energy level
5. $B \rightarrow A$ $\because$ the least amount of energy is emitted in this transition and energy is directly related to frequency.
6. $3.31 \times 10^{-6} \text{ m}$ $\because$ the energy of the absorbed photon is equal to the difference in energy levels and $E = hc/\lambda$
7. 3 $\because$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n + 1)/2$.
8. $B \rightarrow A$ $\because$ the least amount of energy is emitted in this transition and energy is inversely related to wavelength.
9. 2s, 3s, 4s, 5s, 4d
10. $4.10 \times 10^{-7} \text{ m}$ $\because \Delta E = \left(\frac{-R_y}{n_2^2}\right) - \left(\frac{-R_y}{n_1^2}\right)$ and $\Delta E = \frac{hc}{\lambda}$
11. 2s, 3s, 4s, 5s, 4d
12. (lowest frequency) radio waves < microwaves < orange light < green light (highest frequency)
13. $A \rightarrow B$ $\because$ the least amount of energy is absorbed in this transition and energy is inversely related to wavelength.
14. 21 $\because$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n + 1)/2$.
15. 150 zJ $\because$ "A" is the level with the lowest energy, AKA the ground state.
16. 0.63% $\because \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
17. 200 zJ $\because$ "C" is the level with the third-lowest energy, AKA the second excited state.
18. 3.244 m
19. $A \rightarrow G$ $\because$ the most amount of energy is absorbed in this transition and energy is directly related to frequency.
20. 4p: $n = 4, l = 1, N_e = 6$... 3p: $n = 3, l = 1, N_e = 6$... 4d: $n = 4, l = 2, N_e = 10$
21. 120 zJ $\because$ the energy of the absorbed photon is equal to the difference in energy levels. $|150 - 270| = 120$
22. $A \rightarrow B$ $\because$ the least amount of energy is absorbed in this transition and energy is directly related to frequency.
23. 60 zJ $\because$ "B" is the level with the second-lowest energy, AKA the first excited state.
24. (lowest energy) red light < green light < ultraviolet light < gamma rays (highest energy)
25. $G \rightarrow A$ $\because$ the most amount of energy is emitted in this transition and energy is inversely related to wavelength.
26. (shortest wavelength) x-rays < blue light < red light < infrared light (longest wavelength)
27. $6.005 \times 10^{-7} \text{ m}$